王业涵

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教育背景

华东师范大学 **中国，上海**

计算机科学与技术 | 本科 | GPA: 3.71/4.00 **2024.09-至今**

主修课程： 量子计算、数学分析、线性代数、离散数学、数据结构、算法分析与设计、数字逻辑电路、大学物理、概率论与数理统计、操作系统、机器学习等。

交流项目：牛津大学暑期班（师从 Prof. Kobi Kremnizer，修读《Quantum Computing》，成绩 97 分） 2025.08

青年使者赴美（加州）交流项目 **2025.01-2025.02**

实习经历

上汽大众（SAIC VOLKSWAGEN）| 基于 RAG 的外购件质量专家模型 **2026.07-2026.09**

- **返库材料清洗：** 面向多份 PPTX 与 PDF 格式的质量问题单，分别用 python-pptx 与 PyMuPDF 按页抽取正文。再以预定义五字段为 Schema，用 LLM 做约束式改写，将原文归纳为车型、零件、问题概述、根本原因和遏制措施，形成可按车型、零件筛选的案例表。
- **案例库 RAG 部署：** 在结构化案例上构建 FAISS 与 BM25 混合检索，经 RRF 融合与 Reranker 后生成回答，同时开发上传预览入库、删除记录及索引热更新，便于后续问题单进入知识库。
- **模块优化：** 在“任务流程”模块补齐流程绑定、软删除功能和上手指引，降低新人按规程流转的成本。

项目经历

基于 GitHub Pages 的个人主页 | 第一开发人 **2026.08-2026.09**

- **单页应用架构设计：** 基于 Vite + React + TypeScript 构建单页应用，采用长滚动首页与子路由相结合的结构，内容与组件解耦，以强类型 TypeScript schema 管理站点数据。
- **交互式地图与运动数据可视化：** 集成 Leaflet 渲染 GPX 运动轨迹与 GeoJSON 旅行路线，实现运动数据与生活足迹的动态地图展示，支持缩放拖拽与轨迹动态绘制，将原始 GPS 数据转化为可探索的交互式可视化体验。
- **隐私友好的数据分析与自动化部署：** 接入 GoatCounter 实现无 Cookie 访问统计，保护访客隐私；配置 GitHub Actions 工作流，实现代码推送后自动构建部署至 GitHub Pages，支持持续集成与快速迭代。
- 项目已上线，可通过 **https://visionnext100.github.io** 访问，持续维护中。

医疗类 AI 产品 AdenoGuard——儿童腺样体智能筛查与辅助决策系统 | 第一开发人 **2026.01-2026.03**

- **真实临床数据与模型调优：** 依托三甲医院授权的 10GB+ 真实多模态临床数据（既往病史、腔镜影像、鼾声音频），开发多模态 AI 引擎以协助筛查腺样体肿大病症。核心搭载 AutoGluon 训练框架，AUC 精度高达 0.91。
- **端云协同架构与小程序开发：** 攻克跨模态数据流转难题，实现多媒体医疗数据的无缝采集与全链路加密传输；对接后端 FastAPI 接口，保障云端 AI 推理快速响应；使用 SQL 管理用户数据库。
- **医疗级闭环交互设计：** 将诊断结果通过 CLAHE 图像增强技术及梅尔谱图可视化呈现，完成从结构化病史录入到治疗辅助决策的业务闭环。利用 Canvas 渲染含置信度仪表盘、风险趋势图的医疗报告单，赋能无创居家早筛。
- 目前已产出体验版小程序，并申请软件著作权与发明专利。

竞赛经历

2026年 MCM/ICM 美国大学生数学建模竞赛 | 队长兼写作手 **2026.01-2026.02**

- **数据清洗与特征工程：** 针对分数归零、淘汰轮次缺失等问题设计标准化清洗策略，通过分数归一化消除赛季差异，并基于领域知识构造“The Kingmaker Effect”等高解释性特征。
- **混合建模与逆向优化：** 综合运用 XGBoost、随机森林与 SLSQP 优化算法构建混合模型，并结合 SHAP 进行特征归因；构建逆向优化模型重构未知投票数据，并通过敏感性分析验证模型稳健性。

OS 华东区域赛 | 核心队员 **2025.12-2026.01**

- **内核级调度器设计：** 针对传统 MLFQ 易被 CPU 密集型进程“欺骗”的漏洞，基于 uCore OS 实现 Smart MLFQ 调度器；引入 EWMA 实时评估 CPU 利用率，动态识别 I/O 密集型与伪装型进程，兼顾响应速度与抗欺骗能力。
- **系统压测与验证：** 构造多类型进程压力测试，对比 RR/Basic MLFQ/Smart MLFQ 的响应与周转时间，验证 Smart MLFQ 在保持 I/O 低延迟的同时有效抑制欺骗行为。

荣誉奖项

- 2026年 本科生国家奖学金获得者
- 2026年 美国大学生数学建模竞赛（MCM/ICM）二等奖（Honorable Mention）
- 2026年 全国大学生计算机系统能力大赛-操作系统设计赛-华东区域赛-操作系统原理赛道 三等奖
- 2025年 华东师范大学优秀学生

专业技能

- **计算机能力：** C++、Python、SQL、JavaScript、HTML、CSS、微信小程序开发
- **语言能力：** 英语水平良好（高考英语 143 分，CET-6: 568分）
- **办公软件：** 熟练使用 Word、PowerPoint、Excel、飞书

Yehan Wang

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Education

East China Normal University **Shanghai, China**

Bachelor of Computer Science and Technology | GPA: 3.71/4.00 **Sep 2024 – Present**

Relevant Courses: Quantum Computing, Mathematical Analysis, Linear Algebra, Discrete Mathematics, Data Structures, Introduction to Algorithm, Digital Logic Circuits, Physics, Probability and Statistics, Operating Systems, Machine Learning.

Exchange Programs:

·Oxford University Summer Institute – Quantum Computing (Advised by Prof. Kobi Kremizer) | Grade: 97/100 **Aug 2025**

·Youth Envoy Exchange Program – California, USA **Jan 2025 – Feb 2025**

Internship

SAIC VOLKSWAGEN | A RAG-based expert model for the quality of purchased components

Jul 2026 – Sep 2026

- **Return-to-plant data cleaning:** For quality issue files in PPTX and PDF, extract page-level text with python-pptx and PyMuPDF respectively. Then run schema-constrained LLM rewriting against five predefined fields—vehicle, part, problem summary, root cause, and containment—to produce a case table filterable by vehicle and part.
- **Case-base RAG deployment:** Build hybrid FAISS and BM25 retrieval over the structured cases, fuse with RRF and a reranker, then generate answers. Also add upload-and-preview ingest, record deletion, and index hot-reload so later issue sheets can enter the knowledge base.
- **Module enhancement:** On the task-flow module, complete process binding, soft delete, and onboarding guidance to lower the cost for newcomers following standard procedures.

Projects

GitHub Pages personal site | Lead Developer **Aug 2026 – Sep 2026**

- **Single-page architecture:** Built an SPA with Vite, React, and TypeScript, combining a long-scroll home page with nested routes. Decoupled content from components and managed site data with a strongly typed TypeScript schema.
- **Interactive maps and activity visualization:** Integrated Leaflet to render GPX workout tracks and GeoJSON travel routes, with pan, zoom, and animated path drawing, turning raw GPS logs into an explorable map experience.
- **Privacy-friendly analytics and automated deploy:** Added cookieless visit stats via GoatCounter. Configured GitHub Actions so pushes build and deploy to GitHub Pages, supporting CI and fast iteration.
- Live at **https://visionnext100.github.io** and under active maintenance.

AdenoGuard — a pediatric-adenoid AI screening and decision-support product | Lead Developer

Jan 2026 – Mar 2026

- **Clinical data and model tuning:** Used 10GB+ authorized multimodal data from a tertiary hospital—history, endoscopic images, and snore audio — to build a multimodal AI engine for adenoid hypertrophy screening. Trained with AutoGluon; AUC reached 0.91.
- **Device–cloud architecture and mini-program:** Solved cross-modal data flow so multimedia clinical inputs can be captured and sent end-to-end encrypted; connected a FastAPI backend for low-latency cloud inference; managed users with SQL.
- **Closed-loop clinical UX:** Presented findings with CLAHE-enhanced images and mel spectrograms, from structured history intake to decision support, and rendered Canvas reports with a confidence gauge and risk trend for at-home screening.
- Shipped an experience-edition mini-program; software copyright and invention patent applications filed.

Contests

MCM/ICM 2026 | Team Leader & Writer **Jan 2026 – Feb 2026**

- **Data cleaning and feature engineering:** Designed a pipeline for zeroed scores and missing elimination rounds, normalized scores across seasons, and used domain knowledge to build high-interpretability features such as the Kingmaker Effect.
- **Hybrid modeling and inverse optimization:** Combined XGBoost, random forests, and SLSQP, with SHAP for attribution; reconstructed unknown votes by inverse optimization and checked robustness via sensitivity analysis.

CSCC 2025 Operating System Design Competition | Core Team Member **Dec 2025 – Jan 2026**

- **Kernel scheduler design:** Implemented Smart MLFQ on uCore OS to stop CPU-bound jobs gaming classic MLFQ; used EWMA to separate I/O-bound from disguised CPU hogs while keeping response time low.
- **Stress tests and validation:** Built mixed-workload benchmarks and compared RR, basic MLFQ, and Smart MLFQ on response and turnaround time, showing Smart MLFQ keeps I/O latency low while limiting gaming.

Honors & Awards

- National Scholarship for Undergraduates 2026
- MCM/ICM Honorable Mention 2026
- Third Prize – CSCC 2025 Operating System Design Competition (East China Region) 2026
- Outstanding Student Award – East China Normal University 2025

Technical Skills

- Programming: C++, Python, PyTorch, JavaScript, WeChat Mini Program Development
- Languages: Proficient in English (Gaokao: 143/150, CET-6: 568)
- Office Software: Proficient in Word, PowerPoint, Excel, Lark